
EXERCISE

Lecture 9: Routing

AI508 — Algorithmic Game Theory

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1 Contrast to auctions

While actions were mostly based on system design, routing is more based on selfish decisions.

2 Braess paradox

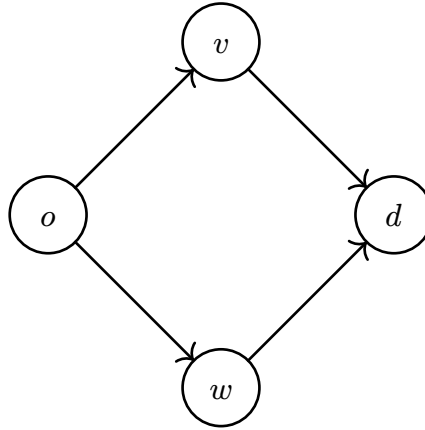


Figure 1: A traffic system

Given that

$$o \rightarrow v : c(x) = x$$

$$o \rightarrow w : c(x) = 1$$

$$v \rightarrow d : c(x) = 1$$

$$w \rightarrow d : c(x) = x$$

where x is the fraction of total users on the road. so $x_{ov} + x_{ow} = 1$

The split should be 50/50 given that going up and down, results in the same cost of $c_T = 1.5$.

Now modify

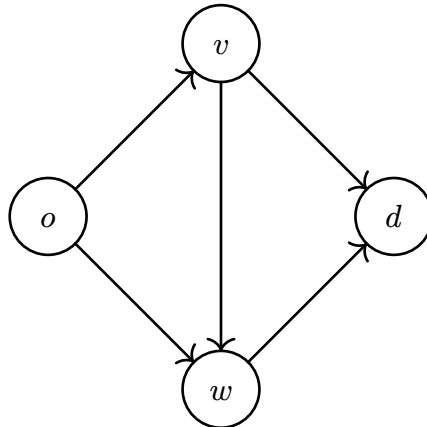


Figure 2: A modified traffic system

And

$$o \rightarrow v : c(x) = x$$

$$o \rightarrow w : c(x) = 1$$

$$v \rightarrow d : c(x) = 1$$

$$w \rightarrow d : c(x) = x$$

$$v \rightarrow w : c(x) = 0$$

Then

$$2x \leq 1 + x$$

Given you've already paid $c = x$ the worst you can do is pay x again, which cannot be worse than 1

Theorem: Price of Anarchy

$$\text{PoA} = \frac{\text{Cost of the system at an equilibrium}}{\text{Best/lowest system performance cost}}$$

3 Pigov Network

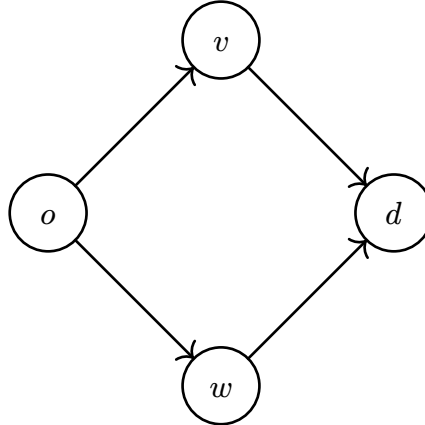


Figure 3: A *very* complicated network

Where

$$o \rightarrow v \rightarrow d : c(x) = 1$$

$$o \rightarrow w \rightarrow d : c(x) = x$$

At equilibrium $x_{\text{owd}} = 1$, and total cost = 1

Then total cost given $x_{\text{split}} = x_{\text{owd}} \cdot x_{\text{owd}} + (1 - x_{\text{owd}}) \cdot 1$

So

$$\frac{d}{dx_{\text{owd}}} x_{\text{split}} = 0 \implies x_{\text{owd}}^* = 1/2$$

Then the best cost for the system is

$$2 \cdot 1/2 + (1 - 1/2) \cdot 1 = 3/4$$

Then the PoA = $\frac{1}{3/4} = 4/3$.

4 supremum notation btw

Consider $c_p = x^p$, where $0 \leq p \leq 1$ Then

$$\text{PoA} = \frac{1}{1 - \frac{p}{(p+1)^{1+1/p}}}$$

As $p \rightarrow 1$, then $\text{PoA} \rightarrow 4/3$

Assume that i want to maximize PoA. Since the max is not defined as $p < 1$.

We use

$$\sup \text{PoA}$$

instead.

5 Generalized Pigov

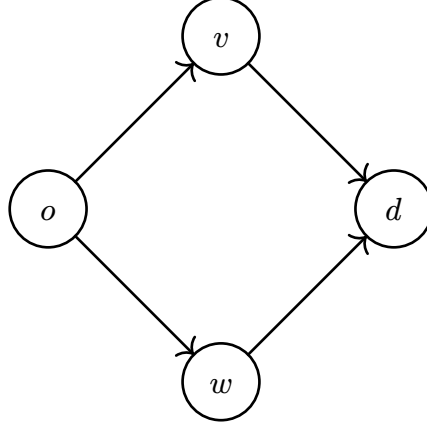


Figure 4: Generalized Pigov

with

$$\begin{aligned} o \rightarrow v \rightarrow d : c(r) & \quad (\text{fixed}) \\ o \rightarrow w \rightarrow d : c(x) & \quad (\text{function}) \end{aligned}$$

Total traffic flow rate $r > 0$, then $0 \leq x \leq r$. Also by design c as continuous and non decreasing, so $c(x) \leq c(r)$.

Define

$$G = \{\text{class of cost functions}\} \quad \text{where } c \in G$$

Where

$$\begin{aligned} \alpha(G) &= \sup_{c \in G} \sup_{r > 0} \frac{r \cdot c(r)}{\min_{0 \leq x \leq r} \{x \cdot c(x) + (r - x) \cdot c(x)\}} \\ &= \sup_{c \in G} \sup_{r > 0} \max_{0 \leq x \leq r} \frac{r \cdot c(r)}{x \cdot c(x) + (r - x) \cdot c(x)} \end{aligned}$$

Consider

$$G_{\text{quad}} = \{ax^2 + bx + d : a, b, d \geq 0\}$$

Let $c(x) = x^2$, and $r = 1$. Then

$$\alpha(G) = \sup_{c \in G} \sup_{r > 0} \frac{1}{\min_{0 \leq x \leq 1} x \cdot (x^2 - 1) + 1}$$

given that $\arg \min_{0 \leq x \leq 1} x \cdot (x^2 - 1) + 1 \implies x^* \frac{1}{\sqrt{3}}$

$$\alpha(G) = \sup_{c \in G} \sup_{r > 0} \frac{1}{-2/3\sqrt{3}} = \sup_{c \in G} \sup_{r > 0} \frac{3\sqrt{3}}{-2}$$

But we wish to show this for a generic quadratic cost function

$$\alpha(G) = \sup_{c \in G} \sup_{r > 0} \frac{r \cdot (ar^2 + br + d)}{\min_{0 \leq x \leq r} x \cdot (ax^2 + bx + d) + (r - x) \cdot (ax^2 + bx + d)}$$

Key Theorem

For every set G of cost functions & for every selfish routing network with

$$c_i \in G \quad \forall i$$

Then

$$\text{PoA} \leq \alpha(G) \quad (\text{despite } \alpha \text{ only defined on Pigov networks})$$